Zhuoyuan Li

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Institute for Functional Intelligent Materials
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Education

2020–2025	**Ph.D.**, School of Mathematical Sciences, Peking University, Beijing, China co-supervisors: Prof. Pingwen Zhang[1], Prof. Bin Dong[2]
2016–2020	**B.Sc.**, School of Mathematics, Sichuan University, Chengdu, China

Working Experience

2025–present	Research Fellow	Institute for Functional Intelligent Materials National University of Singapore, Singapore

Research Interests

- Deep learning for multiscale modeling
- Numerical weather prediction and data assimilation
- Inverse problems and uncertainty quantification

Publications

Google Scholar ORCID

† → Equal contribution; ∗ → Corresponding author(s)

Journal Articles

J1. **Zhuoyuan Li**, Dong*, B. & Zhang*, P. State-observation augmented diffusion model for nonlinear assimilation with unknown dynamics. *Journal of Computational Physics* **539,** 114240. ISSN: 0021-9991. https://doi.org/10.1016/j.jcp.2025.114240 (2025).

J2. **Zhuoyuan Li***, Dong, B. & Zhang, P. Latent assimilation with implicit neural representations for unknown dynamics. *Journal of Computational Physics* **506,** 112953. ISSN: 0021-9991. https://doi.org/10.1016/j.jcp.2024.112953 (2024).

[1] homepage: https://www.math.pku.edu.cn/pzhang/en/
[2] homepage: http://faculty.bicmr.pku.edu.cn/~dongbin/

Peer-reviewed Conference Proceedings

C1. Yang[†], P., Feng[†], Y., Chen[†], Z., Wu, Y. & **Zhuoyuan Li**[*]. *Spend Wisely: Maximizing Post-Training Gains in Iterative Synthetic Data Bootstrapping* in *Advances in Neural Information Processing Systems* **38** (2025), 158602–158639. https://doi.org/10.52202/085713-5299.

Preprints

P1. **Zhuoyuan Li**[†,*], Zhao[†], Y. & Li, M. Ensemble Controlled-Flow Filtering for Implicit Data Assimilation. *arXiv preprint arXiv:2607.12975.* under review. https://arxiv.org/abs/2607.12975 (2026).

P2. **Zhuoyuan Li**[†], Zhu[†], A. & Li[*], Q. Hypothesis-driven construction of mesoscopic dynamics. *arXiv preprint arXiv:2605.16211.* under review. https://arxiv.org/abs/2605.16211 (2026).

P3. Li[†], L., **Zhuoyuan Li**[†], Li[†], S., Zhan, K., Gao[*], H., Chen[*], C. & Yang[*], L. In-context modeling as a retrain-free paradigm for foundation models in computational science. *arXiv preprint arXiv:2604.23098.* under review. https://arxiv.org/abs/2604.23098 (2026).

P4. Wang[†], R., Jiang[†], J., **Zhuoyuan Li**, Liu, T., Wang, Y., Xie[*], M. & Piatkevich[*], K. D. Fluorescent protein-based ticker tapes for multiplexed recordings of transcriptional histories in single cells in culture and in vivo. *bioRxiv.* under review. https://www.biorxiv.org/content/early/2025/09/12/2025.09.08.675004 (2025).

P5. Huang[†], X., **Zhuoyuan Li**[†], Liu, H., Wang, Z., Zhou, H., Dong[*], B. & Hua, B. Learning to simulate partially known spatio-temporal dynamics with trainable difference operators. *arXiv preprint arXiv:2307.14395.* https://arxiv.org/abs/2307.14395 (2023).

Talks and posters

EASIAM 2026 Contributed talk — Aug. 23–27, 2026
Hypothesis-driven construction of mesoscopic dynamics

AI4X 2026 Contributed talk — Jun. 15–19, 2026
Learning non-equilibrium mesoscopic dynamics with Onsager principle

CSIAM 2024 Outstanding poster — Oct. 24–27, 2024
Latent assimilation with implicit neural representations for unknown dynamics

Teaching

Peking University

Advanced Algebra (II) Assistant Instructor — Spring 2023
Advanced Algebra (I) Assistant Instructor — Fall 2022
Advanced Algebra (II) Assistant Instructor — Spring 2022
Advanced Mathematics (C) Teaching Assistant[3] — Fall 2020

Please see my homepage for more details.

[3] without teaching tasks

Last updated: August 7, 2026